Solutions to Q2:

1. How many items in the dataset? There are 5 items {a, b, c, e, f}

2. How many itemsets can be generated? There are 2^5-1 = 31 possible itemsets

3. Please calculate the frequency count of 2-itemset {c,e}. The frequent count of {c,e} is 3.

4. Is itemset {b,c} frequent or not? Yes, because support({b,c})=3/5=0.6>0.4

5. What is the confidence of the association rule: {b} -> {c}? The confidence({b}->{c})=count{b,c}/count{b}=3/3=1

6. If min_conf=40%, is the rule {b} -> {c} strong? Yes, because confidence({b}->{c})=1>0.4 and support({b,c})=3/5=0.6>0.4 (two conditions are satisfied)

Solutions to Q3:

If there are 10 items, how many possible itemsets could be checked? The number of possible itemsets is 2^10-1=1023.

Solutions to Q4:

TID	Items in Transaction
1	Bread, eggs, milk
2	Juice, milk
3	Butter, juice
4	Bread, eggs, milk
5	Coffee, eggs
6	Coffee
7	Coffee, juice
8	Bread, cookies, eggs, milk
9	Butter, cookies
10	Bread, milk

There are two steps to mine association rules:

1. Find frequent k-itemsets (support is no less than *minsup=0.2*)
2. Based on frequent k-itemsets found by 1, generate high confidence rules (confidence is no less than *min_conf=0.8*)

Solution to Frequent Itemsets:

1. Set k=1, 2, 3, and find Cl_1, 2, 3 (candidate 1, 2, 3-itemsets). Note that the total number of transactions is 10.

Cl_1	Support
{Bread}	4/10=0.4
{Butter}	2/10=0.2
{Coffee}	3/10=0.3
{Cookies}	2/10=0.2
{Eggs}	4/10=0.4
{Juice}	3/10=0.3
{Milk}	5/10=0.5

Fl_1
{Bread}
{Butter}
{Coffee}
{Cookies}
{Eggs}
{Juice}
{Milk}

Cl_2	Support
{Bread, butter}	0/10=0
{Bread, coffee}	0/10=0
{Bread, cookies}	1/10=0.1
{Bread, eggs}	3/10=0.3
{Bread, juice}	0/10=0
{Bread, milk}	4/10=0.4
{Butter, coffee}	0/10=0
{Butter, cookies}	1/10=0.1
{Butter, eggs}	0/10=0
{Butter, juice}	1/10=0.1
{Butter, milk}	0/10=0
{Coffee, cookies}	0/10=0
{Coffee, eggs}	1/10=0.1
{Coffee, juice}	1/10=0.1
{Coffee, milk}	0/10=0
{Cookies, eggs}	1/10=0.1
{Cookies, juice}	0/10=0
{Cookies, milk}	1/10=0.1
{Eggs, juice}	0/10=0
{Eggs, milk}	3/10=0.3
{Juice, milk}	1/10=0.1

Fl_2

{Bread, eggs}
{Bread, milk}
{Eggs, milk}

CI_3	Support
{Bread, eggs, milk}	3/10=0.3

FI_3
{Bread, eggs, milk}

2. Based on FI_3, we need to find all candidate 4-itemsets. However, we cannot make 4-itemsets based on FI_3 (only one 3-itemset and three items). In other words, CI_4 is empty. Thus, algorithm stops. **Note that algorithm also stops when Frequent k-itemsets are empty. (For example, we assume that the support of {Bread, butter, eggs} is less than min_sup (0.2), then we cannot find out Frequent 3-itemset. FI_3 will be empty, which also stops algorithm.)
3. In all, we have found all frequent k-itemsets (support ≥0.2):
   FI_1: {Bread}, {Butter}, {Coffee}, {Cookies}, {Eggs}, {Juice}, {Milk}
   FI_2: {Bread, eggs}, {Bread, milk}, {Eggs, milk}
   FI_3: {Bread, eggs, milk}

Solution to Strong Association Rules:

To find out mining rules with high confidence, we shall start the rule generation from k=2 since we cannot find out any association rule when k=1. For example, {Bread} is a frequent 1-itemset, we cannot generate any meaningful association rule using only one item. So we start from k=2:

FI_2: {Bread, eggs}, {Bread, milk}, {Eggs, milk}

All possible rule combinations of FI_2 and corresponding confidence values are as follows:
Note: SC() is support count operation.

Rules	Confidence
{Bread} -> {Eggs}	$=\frac{sc(bread, eggs)}{sc(bread)} = \frac{3}{4} = 0.75$
{Eggs} -> {Bread}	$=\frac{sc(eggs, bread)}{sc(eggs)} = \frac{3}{4} = 0.75$
{Bread} -> {Milk}	$=\frac{sc(bread,milk)}{sc(bread)} = \frac{4}{4} = 1.00$
{Milk} -> {Bread}	$=\frac{sc(milk, bread)}{sc(milk)} = \frac{4}{5} = 0.8$
{Eggs} -> {Milk}	$=\frac{sc(eggs,milk)}{sc(eggs)} = \frac{3}{4} = 0.75$
{Milk} -> {Eggs}	$=\frac{sc(milk, eggs)}{sc(milk)} = \frac{3}{5} = 0.6$

Since the *min_conf*=0.8, we keep {Bread} -> {Milk} and {Milk} -> {Bread} when k=2.

FI_3: {Bread, eggs, milk}

All possible rule combinations of FI_3 and corresponding confidence values are as follows:

Rules	Confidence
{Bread} -> {Eggs, Milk}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(bread)}} = \frac{3}{4} = 0.75$
{Eggs} -> {Bread, Milk}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(eggs)}} = \frac{3}{4} = 0.75$
{Milk} -> {Bread, Eggs}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(milk)}} = \frac{3}{5} = 0.60$
{Bread, Eggs} -> {Milk}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(bread, eggs)}} = \frac{3}{3} = 1.00$
{Bread, Milk} -> {Eggs}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(bread, milk)}} = \frac{3}{4} = 0.75$
{Eggs, Milk} -> {Bread}	$=\frac{\text{sc(bread, eggs, milk)}}{\text{sc(eggs, milk)}} = \frac{3}{3} = 1.00$

Since the min_conf=0.8, we keep {Bread, Eggs} -> {Milk} and {Eggs, Milk} -> {Bread} when k=3.

In all, the strong rules (support ≥minsup (0.2) and confidence ≥minconf (0.8)) are as follows:

{Bread} -> {Milk},

{Milk} -> {Bread},

{Bread, Eggs} -> {Milk},

{Eggs, Milk} -> {Bread}.